



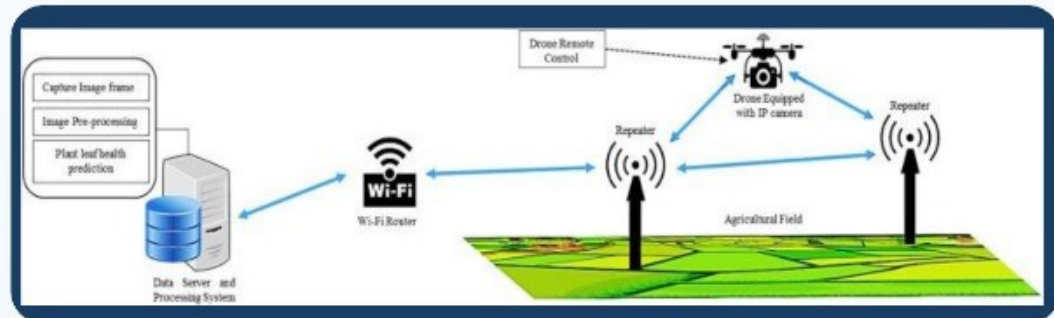
CENTRE FOR ADVANCED STUDIES

LUCKNOW, UTTAR PRADESH, INDIA

INSPIRING STUDENT INNOVATION

Artificial Intelligence-Based Method for Automatic Diagnosis of Chlorosis in Vigna Mungo Plant

A team of Centre for Advanced Studies developed an Artificial Intelligence based method for automatic diagnosis of chlorosis in Vigna Mungo plant. Too early detection of chlorosis problem in vigna mungo plant is possible using this method. The detection is based on a leaf pattern that can use to predict different problem related to the vigna mungo plant which is necessary for proper growth of the plant. The method uses machine learning and deep learning for identifying the severity of chlorosis in vigna mungo plant. The method can be integrated with drones to surveillance the large cultivated fields in real-time. This type of system can be highly beneficial for farmers with large crop fields.



AN INNOVATION BY STUDENTS OF CENTRE FOR ADVANCED STUDIES THAT WILL HELP IN TIMELY IDENTIFICATION OF CHLOROSIS IN VIGNA MUNGO PLANT AND TO GROW CROPS PROPERLY WHICH CAN BE USEFUL FOR FARMERS IN BETTER PRODUCTION OF VIGNA MUNGO PLANTS.

RESEARCH TEAM



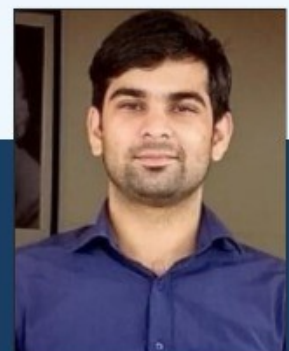
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RESEARCH PUBLICATION FROM THIS WORK

- 1- Rakesh Chandra Joshi, Manoj Kaushik, Malay Kishore Dutta, Ashish Srivastava & Nandlal Choudhary, "VirLeafNet: Automatic Analysis and Viral Disease Diagnosis Using Deep-Learning in Vigna Mungo Plant" Ecological Informatics, doi.org/10.1016/j.ecoinf.2020.101197, 2020, Elsevier Publishers, SCI indexed Impact Factor – 2.51
- 2- Chandrasen Pandey, Neeraj Baghel, Malay Kishore Dutta, Ashish Srivastava & Nandlal Chaudary, "Machine Learning Approach for Automatic Diagnosis of Chlorosis in Vigna Mungo Leaves", Multimedia Tools and Applications, Multimedia Tools and Applications, Springer Nature Publishers, DOI:10.1007/s11042-020-10309-6, 2020, SCI indexed Impact Factor - 2.30.